

Furbinator 3000: This DIY camera could be a nature photographer's new ally

IT expert James Milward has harnessed artificial intelligence to transform an outdoor camera, which he calls Furbinator 3000, into a sentinel for his garden, proficient at identifying and deterring unwanted badger and fox visitors. However, this novel project, initiated as a solution to keep these creatures at bay, has caught the attention of nature photographers, who perceive its potential to revolutionise their craft.

Milward's ingenious endeavour entails custom-built code, closely monitoring a Ring camera's live feed to discern whether the guests are the bothersome creatures he seeks to deter. When the camera spots a badger or fox, it activates an ultrasonic repellent he procured from Amazon. The intriguing prospect here is how this technology can be adapted for photographers to enhance their wildlife photography.

The core of Milward's project centres on training the cameras to distinguish between foxes and badgers, as each requires a different frequency of sound for effective deterrence. By enabling the camera to identify the intruder, the right deterrent can be activated, offering a promising addition to wildlife photographers' toolkits.

To build this innovation, Milward employed a Raspberry Pi 4 and TensorFlow Lite, a renowned machine learning tool. Video clips from his Ring camera were manually tagged as foxes or badgers using the LabelImg tool. This critical step prevented the computer from misidentifying common objects. The model was refined with additional steps by utilising Google Colab.

All in all, this project offers a practical solution for photographers seeking to perfect their craft while coexisting with garden wildlife.

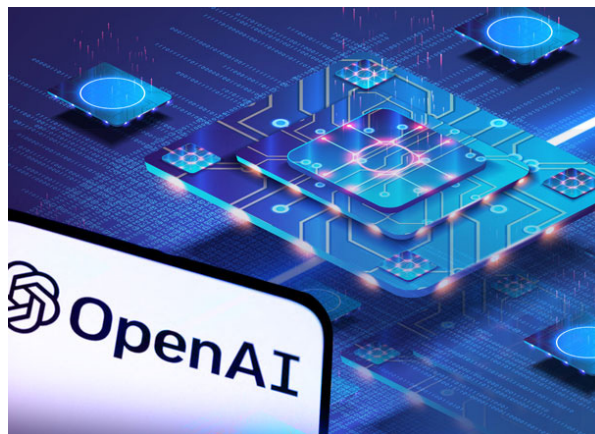
Oligo has collectively termed these vulnerabilities as 'ShellTorch'. Two of these vulnerabilities are rated critical, with near-maximum severity ratings on the CVSS scale. They include CVE-2023-43654, a server-side request forgery (SSRF) vulnerability allowing remote code execution, and CVE-2022-1471, a Java deserialisation RCE flaw.

The third vulnerability arises from TorchServe's default configuration, which exposes a critical management API to the internet. Although modifying this configuration can mitigate the issue, many organisations and projects using TorchServe have retained the default settings.

These vulnerabilities highlight the reality that AI applications face the same risks as other software when it comes to open source code. However, in AI, the consequences can be severe due to its numerous applications.

OpenAI seeks hardware independence

Nvidia currently dominates the AI market, providing the hardware essential for training expansive artificial intelligence (AI) language models. With the rising demand, the availability and prices of their chips have become challenging.



OpenAI, the creator of ChatGPT, is actively exploring alternatives to Nvidia's solutions, including the possibility of developing its proprietary chips.

The company is devising plans to address the current hardware limitations in the AI sector. Among the solutions under

consideration, one radical approach includes acquiring one of the former rivals and granting it the capability to manufacture its chips. So far, the investigation has reached the stage of identifying potential merger candidates.

Sam Altman, OpenAI's CEO, has underscored the urgency of procuring more AI chips, marking it a paramount concern for the company. Yet, fulfilling this need isn't straightforward. Nvidia is grappling with manufacturing constraints, resulting in an inability to meet the demand for its H100 AI chips for an anticipated 1.5 years. The ChatGPT creator previously harnessed 10,000 GPUs to train ChatGPT. Its ambition to scale further is hampered by the scarce supply and skyrocketing prices of chips.

The financial implications are staggering. For the company to handle a query volume equivalent to a tenth of Google's, a whopping \$48 billion investment in graphics processing units (GPUs) would be necessitated. Moreover, maintaining this scaled operation would subsequently demand an annual expenditure of \$16 billion.

The company's contemplation of custom chips mirrors strategies adopted by tech behemoths like Amazon, Google, and Meta, who have bespoke chips tailored to their unique requirements. Microsoft, the OpenAI's principal partner, is reportedly developing its custom silicon. However, its journey to custom silicon realisation will span years. Meanwhile, the reliance on chip giants like Nvidia and AMD remains inevitable.

OpenUK and UN unite for Open Source Data Centre Challenge

In a bid to combat the environmental impact of data centres, OpenUK, representing the UK's open technology sector, has partnered with the United Nations' Office